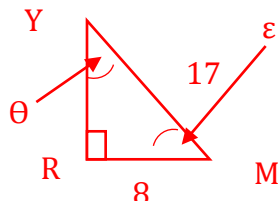


Geometry  
Unit 7  
Lesson 5 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

$\triangle YRM$  is a right triangle where  $m\angle YRM = 90^\circ$ ,  $RM = 8$ , and  $YM = 17$ .

1. Determine  $m\angle YMR$  using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.



$$\sin(\theta) = \frac{\text{opp.}}{\text{hyp.}} = \frac{8}{17}$$

$$\theta = \sin^{-1}\left(\frac{8}{17}\right) = 28.07^\circ$$

$$90^\circ + \theta + \epsilon = 180^\circ$$

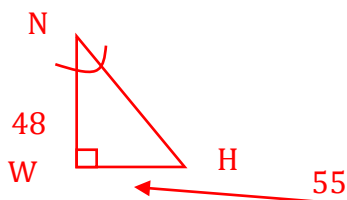
$$\therefore \epsilon = 180^\circ - (90^\circ + 28.07^\circ)$$

$$= 61.93^\circ$$

$$m\angle YMR = 61.928^\circ$$

$\triangle WHN$  is a right triangle where  $m\angle NWH = 90^\circ$ ,  $NW = 48$ , and  $WH = 55$ .

2. Determine  $m\angle WNH$  using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

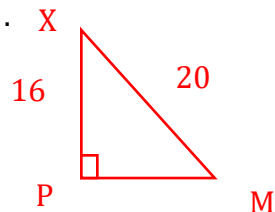


$$\tan \theta = \frac{55}{48}$$

$$\tan^{-1}\left(\frac{55}{48}\right) = 48.888^\circ$$

$\triangle XPM$  is a right triangle where  $m\angle XPM = 90^\circ$ ,  $XP = 16$ , and  $XM = 20$ .

3. Determine the length of  $\overline{MP}$ . If necessary, round your answer to the nearest thousandth.



$$16^2 + b^2 = 20^2$$

$$256 + b^2 = 400$$

$$b^2 = 144$$

$$b = 12 \quad \overline{MP} = 12$$

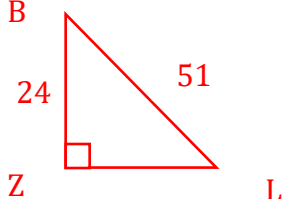
4. Determine  $m\angle PMX$  using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{16}{12}$$

$$\tan^{-1}\left(\frac{16}{12}\right) = 53.130^\circ$$

$\triangle LBZ$  is a right triangle where  $m\angle BZL = 90^\circ$ ,  $BZ = 24$ , and  $LB = 51$ .

5. Determine the length of  $\overline{ZL}$ . If necessary, round your answer to the nearest thousandth.



$$24^2 + b^2 = 51^2$$

$$576 + b^2 = 2601$$

$$b^2 = 2025$$

$$b = 45 \quad \overline{ZL} = 45$$

6. Determine  $m\angle LBZ$  using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{45}{24}$$

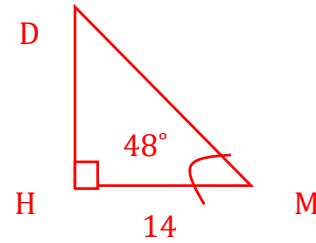
$$\tan^{-1}\left(\frac{45}{24}\right) = 61.928^\circ$$

$\triangle MDH$  is a right triangle where  $m\angle DHM = 90^\circ$ ,  $m\angle DMH = 48^\circ$ , and  $HM = 14$ .

7. Determine the length of  $\overline{DH}$  using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan(48^\circ) = \frac{\overline{DH}}{14}$$

$$\therefore \overline{DH} = 14 \cdot \tan(48^\circ) = \boxed{15.549}$$

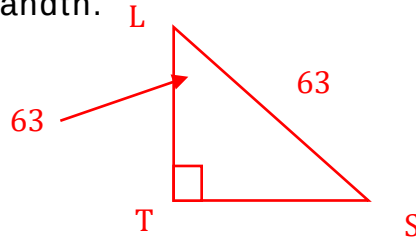


8. Explain why neither the cosine nor the sine ratio could be used to determine the length of  $\overline{DH}$ .

In order to use sine or cosine rate, the length of the hypotenuse of the right triangle must be given. Since the length of the hypotenuse is not provided, the only ratio that is possible to use based on the given information is tangent.

$\triangle TLS$  is a right triangle where  $m\angle LTS = 90^\circ$ ,  $m\angle TLS = 63^\circ$ , and  $LS = 63$ .

9. Determine the length of  $\overline{TL}$ . Show your work. If necessary, round your answer to the nearest thousandth.



$$\cos(63) = \frac{x}{63}$$

$$28.601 = x$$

10. Determine the length of  $\overline{TS}$ . Show your work. If necessary, round your answer to the nearest thousandth.

$$28.601^2 + x^2 = 63^2$$

$$x^2 = 3150.983$$

$$x = 56.134$$